

Figure 2: Difference between generative AI and traditional AI

to the general public. These interfaces mark a substantial advancement in natural language processing (NLP) and artificial intelligence. Other market leaders are Google's PaLM models and Meta's Llama models, which are bidirectional encoder representations from transformers.

Owing to the extensive volume of data needed to train them, LLMs are designed to understand and produce writing like a person, among other types of content. They can translate into languages other than English, infer from context, produce coherent and contextually relevant responses, summarise content, respond to queries (both general and FAQ-related), and even help with creative writing or code-generating activities. They can also be used for writing code, or translating between programming languages.

Not all generative AI tools are built on LLMs, but all LLMs are a form of generative AI. Generative AI is a broad category for a type of AI, referring to any artificial intelligence that can create original content. LLMs are the text-generating part of generative AI.

LLM models can perform various business and personal tasks when trained and given appropriate prompts. Figure 3 shows some of the popular use cases.

## Understanding LLMs from a developer's perspective

Artificial intelligence has undergone a revolution because of LLMs. These cutting-edge AI models are revolutionising how humans engage with technology by enabling the development of robust software applications through APIs. For developers, LLMs provide a plethora of new features, and come in two main categories:

- Base LLMs
- Instruction-tuned LLMs

A language model that creates text based on the input it gets without clear instructions or direction is commonly referred to as a 'base LLM'. With the help of text training data, which is frequently gathered from a variety of sources, including the internet, base LLM is trained to anticipate the next word based on the most likely word that will follow. On the other hand, a language model that receives explicit instructions or assistance during the text creation process is referred to as 'instruction-tuned' LLM. This can entail imposing strict constraints or guidelines to affect the model's results. Instruction-tuned

LLM training often begins with a basic LLM that has been trained on a large volume of text data. Afterwards, inputs and outputs consisting of instructions and sincere attempts to comply with them are used to further train and fine-tune the system.

To make the system even more beneficial and instruction-tuned, it is frequently improved through the use of a method known as reinforcement learning from human feedback, or RLHF. Because instruction-tuned LLMs have been shown to be helpful, honest, and harmless, many real-world usage scenarios have switched to them. Compared to basic LLMs, they are less likely to produce harmful outputs.

## Tuning LLMs

The most common way to interact with the LLM while creating and testing prompts is through an API. Tweaking these settings is important to improve the reliability and desirability of responses. The typical configurations or tuning you would encounter while utilising various LLM providers are described here.

**Temperature:** For tasks like fact-based QA, you might want to use a lower temperature value to encourage more factual responses.

**top\_t:** A sampling technique with temperature, called nucleus sampling, allows you to control how deterministic the model is. Increasing top\_t to a higher value will create a more diverse response. The general recommendation

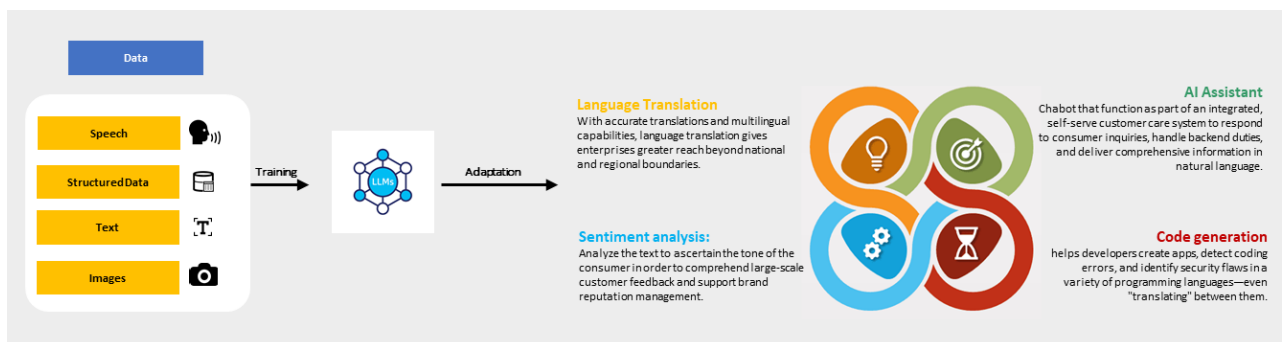


Figure 3: Popular use cases